

Impact of Sexual and Reproductive Health Education on Adolescent Well-being

Aditi Choudhary

John Munro Godfrey, Sr. Department of Economics
University of Georgia

Abstract

This paper evaluates the causal impact of the Peer Educator initiative under the Rashtriya Kishor Swasthya Karyakram (RKSK) on adolescents' knowledge of HIV, teenage pregnancy, and attitudes toward gender-based violence in India. Implemented in selected High Priority Districts (HPDs) across each state, the program allows for a quasi-experimental analysis using a difference-in-differences framework. The results indicate that the program significantly increased HIV awareness by 11.7 percentage points and the use of modern contraception by 2.33 percentage points. However, it did not lead to significant improvements in comprehensive knowledge of HIV or attitudes toward gender-based violence.

(JEL I18, J13, J16, O15)

1 Introduction

India currently faces a significant public health challenge, with over 2.5 million people living with HIV, the second-highest number globally. Alarming, data from a nationally representative survey (2019-21)¹ show that only 20% of young women and 29% of young men aged 15-24 possess comprehensive knowledge of HIV, a decline from the last round of survey conducted in 2015-16². This lack of awareness is particularly concerning given that India's adolescent population exceeds 253 million³, accounting for over one-fifth of the total population. As the government reaffirms its commitment to achieving the Sustainable Development Goal of ending HIV/AIDS as a public threat by 2030, addressing these knowledge gaps becomes essential.

Comprehensive sexual and reproductive health education is essential not only for improving health outcomes but also for addressing the broader issues of gender inequality (Jayachandran, 2015). Adolescence is a critical period for shaping traits that affect future autonomy (Heckman and Mosso, 2014). By equipping adolescents with the necessary knowledge and skills, these interventions can contribute to long-term social and economic benefits. Specifically for girls, it could help them make informed choices, improve their career prospects and life satisfaction, and potentially strengthen their bargaining power within the household (Bandiera et al., 2020).

Given this, exploring effective methods to provide sexual and reproductive health (SRH) education to adolescents is crucial for designing interventions and policies that shape their knowledge, attitudes, and behaviors effectively. One such initiative is the Peer Educator program under the Rashtriya Kishor Swasthya Karyakram (RKSK) program in India, it aims to provide comprehensive SRH education and services to young people. The program seeks to empower and improve the well-being of adolescents in India through community-based initiatives. In this program, trained peers engage with their fellow adolescents to impart essential health knowledge and promote healthy behaviors. This paper seeks to address the following question: How does sexual and reproductive health education through peer education impact the knowledge of HIV, teenage pregnancy, and the overall well-being of adolescents in India?

My empirical strategy assesses the causal impact of the Peer Educator program under the RKSK by comparing treated and control groups. The treated group consists of High Priority Districts (HPDs)⁴, which are identified by the government as having the most critical health challenges based on a composite health index. These districts represent the bottom 25% of districts in each state according to this ranking. Within these HPDs, the program was implemented in 50% of the blocks⁵. The control group comprises the remaining districts in the state. Using a difference-in-differences framework, I estimate the program's effects on various outcomes, including knowledge of sexually transmitted diseases (STDs), teenage pregnancies, and attitudes toward gender-based violence. I use the data from multiple rounds of representative survey data, the National Family Health Survey of India (NFHS-1,2,4,5). Specifically, women aged 15 to 25 years were selected for the analysis.

The results indicate that while the program effectively increases general awareness of HIV, it falls short of enhancing comprehensive knowledge. Specifically, being in a treated district raises the likelihood of having heard of HIV by 11.7 percentage points, though the effect on comprehensive knowledge is not statistically significant. The program also improves both knowledge and use of modern contraception, with awareness increasing by 1.3 percentage points and contraceptive use rising by 2.33 percentage points among adolescents in treated districts, respectively.

In terms of teenage pregnancy, the program is associated with a delayed age at first birth and a 2.4 percentage point decline in teenage pregnancy rates. Additionally, while there is a decrease in the percentage of women who believe wife-beating is justified, these results are not statistically significant. Overall, the results suggest that the program has been effective in increasing general awareness of HIV and improving contraceptive use, but it has not yet made a significant impact on comprehensive HIV knowledge or attitudes toward gender-based violence.

¹National Family Health Survey (NFHS-5)

²NFHS-4

³Census 2011

⁴District: equivalent to a county in the USA

⁵Local unit of administrative division, next to District

This paper contributes to the literature on HIV prevention policies in developing countries. Previous research, such as Dupas (2011), has shown that providing adolescents with detailed risk information can lead to significant behavioral changes, such as a 28% decrease in teen pregnancy. Similarly, Duflo et al. (2015) find that education subsidies in Kenya reduced dropout rates and early pregnancies but did not significantly impact sexually transmitted infections (STIs), suggesting that more comprehensive approaches might be necessary. Furthermore, Dupas et al. (2018), find that both risk information and risk salience improved health knowledge and reduced teen pregnancy rates in Cameroon. This paper contributes to this body of work by assessing how a government program implemented at scale can influence adolescent sexual behavior and health outcomes, thereby expanding the evidence base on the effectiveness and scalability of HIV prevention strategies. The HIV prevention literature has largely focused on randomized controlled trials (RCTs) evaluating specific interventions. Unlike these studies, which often assess targeted programs, this research examines the effectiveness of a large-scale state-implemented sexual and reproductive health education initiative in India. By evaluating a broad, government-led policy, this study provides insights into how such extensive programs can impact adolescent behavior at a national level.

This study also contributes to the growing body of research on empowering adolescent girls in developing countries, particularly through the provision of sexual and reproductive health (SRH) education. The existing literature demonstrates that multifaceted interventions combining vocational training, life skills, and shifts in gender-attitude can significantly enhance the agency of adolescent girls. For example, Bandiera et al. (2020) demonstrated that in Uganda, combining vocational training with SRH information led to substantial improvements in both economic and social empowerment. Similarly, Edmonds et al. (2023) find that life skills programs in Rajasthan, India, effectively reduced school dropout rates by addressing gender norms and fostering aspirations. My paper builds on previous research, but focuses specifically on how SRH education can influence adolescent girls' agency, providing a clearer understanding of the impact of targeted information.

Furthermore, this research aligns with Dhar et al. (2022), who explore how gender-attitude interventions in India resulted in changes in attitudes and behaviors, and Buchmann et al. (2018), who examine the effects of empowerment programs and financial incentives in Bangladesh. By investigating whether SRH education alone can achieve similar outcomes—such as delaying early marriage and improving overall well-being, this study deepens the existing literature and underscores the significance of SRH education as a critical component in the broader strategy to empower adolescent girls.

Finally, this paper contributes to the literature on peer effects among adolescents, particularly in the context of empowerment interventions. Existing research highlights the significant role peers play in shaping various outcomes, including academic performance and social behavior. For instance, Dewan et al. (2024) provide evidence from high schools in India showing that a higher proportion of female classmates positively affects both male and female students' test scores. The improved classroom environment, rather than direct achievement spillovers, is suggested as a mechanism behind these effects. Similarly, Dishion and Tipsord (2011) discuss peer contagion, revealing that adolescents' interactions with peers can amplify problem behaviors such as aggression and drug use, while positive peer influences can foster social and emotional development. This research shows that peer effects are crucial for understanding and boosting adolescent empowerment and highlights the potential of using peer dynamics in targeted interventions to improve outcomes. Hence, my study contributes to this literature as it helps understand the role and effectiveness of peer educators in shaping gender norms in a developing country.

2 About the Program

The Rashtriya Kishor Swasthya Karyakram (RKSK) program, initiated by the Government of India in 2014, is a broad initiative focused on adolescents across the country. Its primary objective is to enhance the skills and capacities of adolescent girls and boys to effectively manage their health concerns. The program provides access to a wide range of

services including information dissemination, counseling, preventive measures, curative treatments, and referral services, including legal aid when necessary.

Key components of RKSK involve both community-based and facility-based interventions. Community-based efforts focus on initiatives such as peer education, where trained peers engage with their fellow adolescents to impart essential health knowledge and promote healthy behaviors. Additionally, counselors conduct outreach activities to provide support and guidance to adolescents in need. Meanwhile, facility-based interventions establish adolescent-friendly health clinics across various levels of healthcare delivery. These clinics serve as accessible hubs where adolescents can seek a spectrum of services in a welcoming and supportive environment.

The primary focus of this study centers around the Peer Educator program, specifically targeting its impact through awareness generation. To effectively assess the impact of the Peer Educator program, it is crucial to isolate its effects from the broader bundle of interventions within the program. This study concentrates on the specific informational outcomes such as knowledge of HIV, attitude towards GBV as these outcomes could be uniquely influenced by the Peer Educator program and are not affected by other components of the RKSK program. By focusing on these direct effects, I can better understand the unique contributions of peer education without conflating them with other components of the program.⁶

2.1 Peer Educator Program

The Peer Educator program (PE) under the Rashtriya Kishor Swasthya Karyakram (RKSK) is a key initiative designed to empower adolescents to tackle health challenges in their communities. Under this program, four Saathiya⁷ (two boys and two girls) are selected per village who then undergo specialized training facilitated by Accredited Social Health Activist (ASHA) focusing on peer education techniques and health sensitization.

Criteria for selecting Peer Educators are stringent, including age considerations (15 to 18 years for out-of-school adolescents and 14 to 16 years for school-going adolescents), willingness to commit adequate time to the program, and alignment with the social and demographic characteristics of the target adolescent groups ensuring comprehensive coverage of both in-school and out-of-school adolescents.

2.1.1 Peer Educator Training Program

The Peer Educator training program, facilitated by an ASHA worker, spans six days and includes six modules. Five modules focus on adolescent health, while one enhances leadership skills and teach recording and reporting of PE sessions.

Modules on Adolescent Health

1. Growing Up: Understand the physical and psycho-social changes in adolescence, learn about menstrual cycles, social practices, myths and effective management, and address nightfall in boys and related social practices.

⁶Here is a brief description of the other components of the program:

The Adolescent Friendly Health Clinics (AFHCs) are specialized healthcare facilities established within existing public health institutions across India. These clinics provide a range of youth-friendly services tailored to the specific needs of adolescents, including counseling, preventive measures, curative services, and information on various health topics such as nutrition, contraceptive, RTI/STI, violence and other mental health issues.

The Weekly Iron Folic Acid Supplementation program (WIFS) is a nationwide initiative aimed at addressing nutritional deficiencies, particularly among adolescent girls. School-going adolescent boys and girls, as well as out-of-school adolescent girls, receive weekly iron and folic acid supplements. The supplementation aims to prevent anemia and improve overall health and well-being among adolescents, especially girls who are at higher risk of iron deficiency due to menstruation.

The Menstrual Hygiene Scheme (MHS) focuses on promoting menstrual hygiene practices among adolescent girls. Under this scheme, funds are provided to states and union territories to procure sanitary napkins for distribution to adolescent girls aged 10-19 years at a subsidized rate. By ensuring access to quality menstrual hygiene products, the MHS aims to improve menstrual health and hygiene practices among adolescent girls.

Adolescent Health and Wellness Days (AHWDs) are periodic events organized to raise awareness about adolescent health issues and provide essential health services to young people. AHWDs serve as platforms for community engagement and empowerment, encouraging adolescents to take proactive steps towards maintaining their health and well-being.

⁷Hindi word for a friend

2. Nutrition and Health: Increase awareness about food and nutrition.
3. Healthy Mind and Healthy Behaviors: Recognize substance misuse risks, ways to manage peer pressure assertively. Understand adolescent stress effects.
4. Adolescent Sexual and Reproductive Health: Know about the rights related to legal age for marriage and child marriage consequences, access resources for rights violations. Recognize RTI and STI symptoms and seek timely help.
5. Understanding Gender and Violence: Differentiate between sex and gender, learn about fundamental rights and Gender Based Violence(GBV), prevent GBV and seek help if facing abuse.

After this training, the peer educators serve as dedicated volunteers equipped with leadership qualities and a commitment to the well-being of their peers. They form groups of 15-20 adolescents from their community and conduct weekly interactive sessions using the Peer Education kit (PE kit), comprising activity books and informative materials. Additionally, they play a crucial role in mobilizing adolescents to participate in quarterly Adolescent Health and Wellness Days and refer them to appropriate health services when needed.

Male and female peer educators collaborate to establish Adolescent Health Clubs fostering peer support and community engagement. They maintain detailed diaries documenting session overviews and participant numbers, ensuring accountability and facilitating program evaluation. Overall, the Peer Educator program embodies the spirit of empowerment and community-driven action, harnessing the potential of adolescents to promote health and well-being among their peers.

2.1.2 Implementation

Figure 1 presents a detailed organizational framework for the program, highlighting the administrative divisions and key personnel responsible for its implementation. Managed by the Ministry of Health and Family Welfare, the program is overseen by designated officers at various administrative levels. These officers ensure that the program aligns with national health objectives while being tailored to meet the specific needs of different regions.

At the community level, the program relies significantly on the involvement of Accredited Social Health Activists (ASHA) workers, who serve as village-level Peer Educator (PE) coordinators. ASHAs are essential to the program's success, ensuring that peer education activities are effectively conducted within their communities. They organize meetings, facilitate peer education sessions, and maintain regular communication with both the community and higher-level authorities to ensure the program remains on track. Their role as the primary points of contact between the program and the communities they serve is vital.

Auxiliary Nurse Midwives (ANMs) also play a key role in moderating monthly Adolescent Friendly Counseling (AFC) sessions. They provide essential health education and address the specific health concerns of adolescents. The program's implementation is further demonstrated by the substantial number of adolescents reached and peer educators trained across various states. For example, in Assam, 12,784 Peer Educators (PEs) have been selected out of a target of 13,524 across seven districts, with 10,608 already trained, reflecting a 92% training completion rate. Notably, districts like Nagaon and Karimganj achieved 100% training completion, underscoring the effectiveness of the training sessions.

In Uttar Pradesh, 90,646 PEs have been selected across 25 districts, with 44,514 already trained. Reports from the i-Saathiya Study further highlight the implementation of the Peer Educator Program (PEP) across districts in Madhya Pradesh (MP) and Maharashtra. In Madhya Pradesh, 1,480 adolescents have been enrolled under the NGO-led model, with 104 PEs actively engaged across two districts. Similarly, Maharashtra's state-led model has reached 1,643 adolescents, with 134 PEs in the two districts.

These statistics not only provide an overview of the program's implementation across different states, but they also

illustrate the varying structures of implementation, with both state-led and NGO-led models contributing to the program's reach and effectiveness.

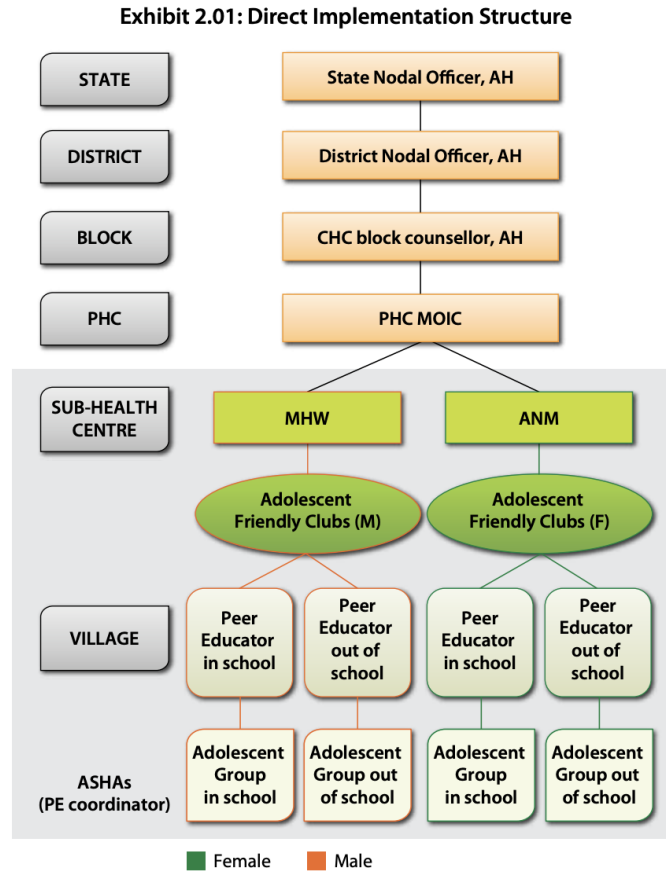


Figure 1: Source: Operational Framework RKSK

- Why and how Peer Educator program can help?

By using trained peers, the program creates a supportive learning environment for adolescents. This approach helps overcome fears or barriers to accessing health services, making it easier for young people to get the care they need. It also helps build information and support networks within their communities, improving access to reliable guidance. Through focused education, the program aims to boost knowledge, attitudes, and behaviors about sexual and reproductive health, reduce teenage pregnancies, and foster more positive views on gender-based violence.

3 Data Description

For this study, I use the data from the 1992-93 to 2019–2021 rounds of the National Family Health Survey of India (NFHS-1,2,4,5)⁸. The datasets collect comprehensive information for women aged 15 to 49, covering various aspects such as reproductive health, sexual behavior, HIV/AIDS, domestic and spousal violence, empowerment, and awareness regarding contraception.

The National Family Health Survey (NFHS) is a nationally representative cross-sectional survey conducted in India, following the guidelines of the Demographic Health Surveys (DHS) conducted worldwide.⁹

⁸NFHS-1 - 1992-93, NFHS-2- 1998-99, NFHS-4 - 2015-16, NFHS-5 - 2019-21 NFHS-3 does not have district information hence I have not used that in the sample

⁹It adopts a stratified sample design, where Primary Sampling Units (PSUs), representing villages in rural areas and Census Enumeration Blocks (CEBs) in urban areas, are selected in the first stage with probability proportional to size (PPS). In the second stage, an equal number of households are randomly selected within each PSU.

The sample for this analysis spans the years 1992 to 2021, capturing a comprehensive time-frame to assess the effects of the program. Specifically, women aged 15 to 25 years were selected for the analysis. Given the program’s emphasis on adolescents, targeting individuals up to the age of 25 ensures the inclusion of those who would have been within the program’s target age range during its implementation. For instance, individuals who received treatment at the age of 19 in 2016 would be 24 years old by 2021, making them eligible for inclusion in the analysis. Hence, 15-25 sample represents the primary beneficiaries of the program.

4 Empirical Strategy

The aim of my empirical strategy is to assess the causal impact of the Peer Educator program within the Rashtriya Kishor Swasthya Karyakram (RKSK). The RKSK’s Peer Educator program has been rolled out in a phased manner, initially covering High Priority Districts (HPDs), with the number of selected districts at 200 by 2020. This program is being implemented in select districts identified as HPDs, and within, these districts 50% of blocks are covered. The government plans to initially saturate all blocks in the selected Peer Educator districts before gradually expanding to other districts based on proposals received from states in their program Implementation Plans(PIPs). I leverage the quasi-experimental design inherent in the implementation of the program to estimate the effect of sexual reproductive health education on various outcomes, including knowledge of sexually transmitted diseases (STDs), early marriage, teenage pregnancies, and attitudes toward gender-based violence.

High Priority Districts: Under the National Health Mission (NHM), the Central Government has designated certain districts in each state as High Priority Districts (HPDs), including aspirational districts identified by NITI Aayog¹⁰. These HPDs represent the bottom 25% of districts in each state based on a composite health index ranking.¹¹ The composite health index¹² is constructed using various health indicators, including – Neonatal Mortality Rate (NMR), Under-five Mortality Rate (U5MR), Total Fertility Rate (TFR), Proportion of Low Birth Weight (LBW) among newborns, Sex Ratio at Birth (SRB), full immunization coverage, proportion of institutional deliveries, total case notification rate of tuberculosis, treatment success rate of new microbiologically confirmed tuberculosis (TB) cases, Proportion of people living with HIV (PLHIV) on antiretroviral therapy (ART), along with indicators from governance and health systems/service delivery domains.

For this study, High Priority Districts (HPDs) included in the RKSK Peer Educator program are considered the treated group, while the remaining districts in the state serve as the control group. The states are responsible for implementing the program according to the operational framework provided by the Ministry of Health and Family Welfare. There are some deviations in the districts chosen for the Peer Educator program from the HPD ranking for some states. For this analysis, I have selected data from states where Peer Educator districts align with the HPD categorization.¹³ I employ a difference-in-differences (diff-in-diff) framework to estimate the coefficient of interest.

For a person i in district d in state s and year t , I adopt the following diff-in-diff estimation framework:

$$Y_{idst} = \alpha + \beta_1 * Treatment_{ds} + \beta_2 Treatment_{ds} * Post_t + \gamma X_{idst} + \delta_s + \theta_t + \epsilon_{idst}$$

where Y_{idst} is the outcome of interest, such as an indicator for knowledge of HIV or teenage pregnancy; $Treatment$ is an indicator for residence in treated district; $Post_t$ equals one if $t > 2017$ and is zero otherwise; and X_{idst} is a vector of covariates that includes religion, region and age. The specification also includes fixed effects for state (δ_s) and year (θ_t).

¹⁰These districts come under Aspirational Districts Programme (ADP) that aims to rapidly and effectively transform the 112 most under-developed districts across the country.

¹¹All Left Wing Affected districts and districts with majority tribal population, whose composite health index is below 50% are also categorized as HPDs

¹²I only have access to list of districts that are high priority and not the ranking of all districts within a state

¹³The analysis includes data from 12 states and 98 treated districts

The coefficient of interest, β_2 , measures the effect of Peer Educator program on the outcome of interest for individuals in treated districts relative to control districts before and after 2017.¹⁴ For all the regressions in the analysis, standard errors are clustered at the district level.

Post period: The post-treatment period starts from 2017 because the assignment of the peer educators and their meetings began after 2016 in most states. The period from 2014 to 2016 was dedicated to infrastructure building and training for the Peer Educator program.

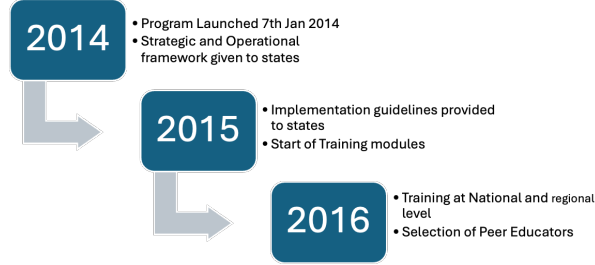


Figure 2: Program Implementation Timeline

Identifying assumption: The diff-in-diff approach assumes that, in the absence of the Peer Educator program, the outcomes of a person in treated districts would have followed the same trend as a similar person in control districts. Next, the summary statistics will provide an overview of the outcome variable and the sample characteristics of both groups while the results section will include an analysis of pre-trends and robustness checks to validate this identifying assumption.

5 Summary Statistics

Table 1 provides the summary statistics of the outcomes and background characteristics in our analysis. It compares the treatment and control groups before and after an intervention across various metrics. Column 1 provide the means for sample with all NFHS rounds up to 2016 as pre-treatment period whereas column 2 gives the means for the NFHS-4 the round right before the program was implemented.¹⁵ In both groups, there was a significant increase in the use and knowledge of modern contraceptive methods, with the treatment group showing a higher improvement, 1.72 and 4.85 percentage points for knowledge and use, respectively. Awareness of HIV/AIDs also increased notably in both groups, with the treatment group experiencing a larger boost in HIV awareness. The comprehensive knowledge of HIV decreased in both groups. Here, comprehensive Knowledge of HIV, constructed from responses to five questions related to HIV transmission and misconceptions.¹⁶ . Additionally, there was a decline in the percentage of births occurring under the age of 19 in both groups, with the treatment group showing a slightly larger reduction.

¹⁴Standard errors clustered at district level

¹⁵The program launched in 2014 however the peer meetings did not start till 2017 hence NFHS 4 is considered as pre-treatment

¹⁶These include: "Can you reduce the risk of getting HIV by always using condoms during sex?" "Can you reduce the risk of getting HIV by having only one sex partner with no other partners?" "Can you get HIV from mosquito bites?" "Can you get HIV by sharing food with a person who has AIDS?" and "Can a healthy-looking person have HIV?" The outcome variable is set to 1 if all questions are answered correctly.

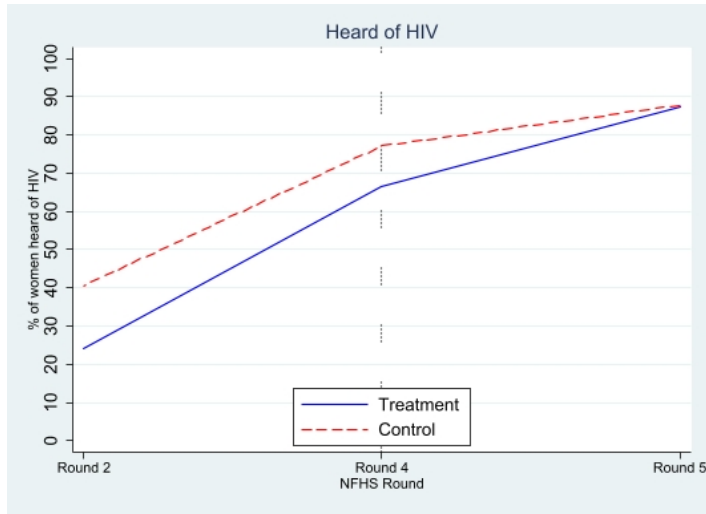
Outcome Variable	Treatment				Control			
	Pre NFHS-1,2,4	Pre NFHS-4	Post NFHS-5	% change col.(3-1)	Pre NFHS-1,2,4	Pre NFHS-4	Post NFHS-5	% change col.(3-1)
Heard of AIDs	52.95	66.38	87.38	34.43	64.10	77.12	87.84	23.74
Comprehensive Knowledge of HIV		28.41	21.29			29.11	21.16	
Knows Modern Methods	95.83	96.16	97.55	1.72	96.95	97.30	97.45	0.50
Uses Modern Methods	8.37	8.04	13.22	4.85	10.72	10.04	13.04	2.32
% Age of Birth less than 19	57.80	52.22	47.70	-10.1	55.55	48.06	46.74	-8.81
Attitude towards GBV		45.79	38.48	-7.31		43.39	35.55	-7.84
Religion (%)								
Hindu	73.83	72.19	75.24	1.41	81.18	80.70	83.11	1.93
Muslim	14.80	15.38	14.60	-0.20	12.25	12.83	10.92	-1.33
Christian	6.57	7.31	4.56	-2.01	2.51	2.49	2.04	-0.58
Sikh	2.64	2.68	3.16	0.72	3.36	3.23	3.31	0.29
Education (%)								
No Education	23.10	16.77	11.09	-12.01	17.87	11.08	6.62	-11.25
Primary	12.14	11.81	9.49	-2.65	10.78	9.58	6.73	-4.05
Secondary	53.54	58.99	65.08	11.54	56.89	63.11	66.71	9.82
Higher	11.19	12.42	14.34	3.15	14.43	16.22	19.94	5.51
Type of Residence (%)								
Urban	20.27	21.09	15.31	-4.96	27.98	28.88	21.61	-6.37
Rural	79.73	78.91	84.69	4.96	72.02	71.12	78.39	6.37

Table 1: Comparison of Variables Between Treatment and Control Groups

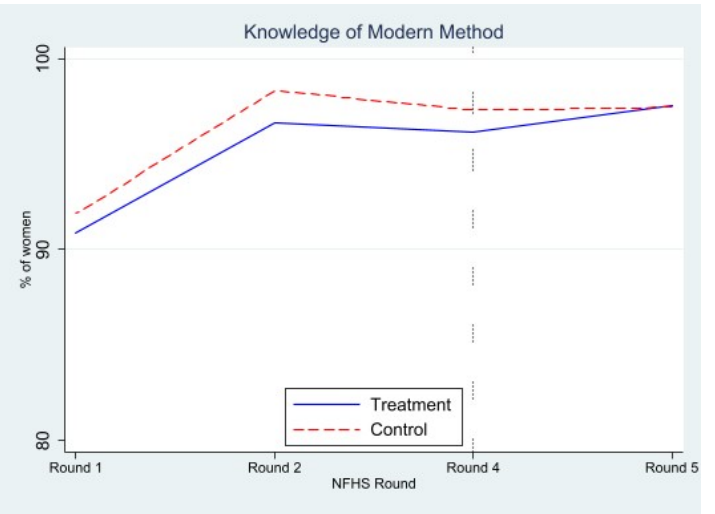
In terms of demographic factors, the religious composition remained relatively stable, with Hindus being the majority, followed by Muslims, Christians, and Sikhs. Education levels improved in both groups, particularly with a drop in the percentage of individuals with no education and an increase in secondary and higher education levels. Rural residence increased slightly in both groups, but more notably in the control group. Overall, the intervention appears to have had a positive impact on modern methods usage, HIV/AIDS knowledge, and reducing early births.

6 Parallel Trends

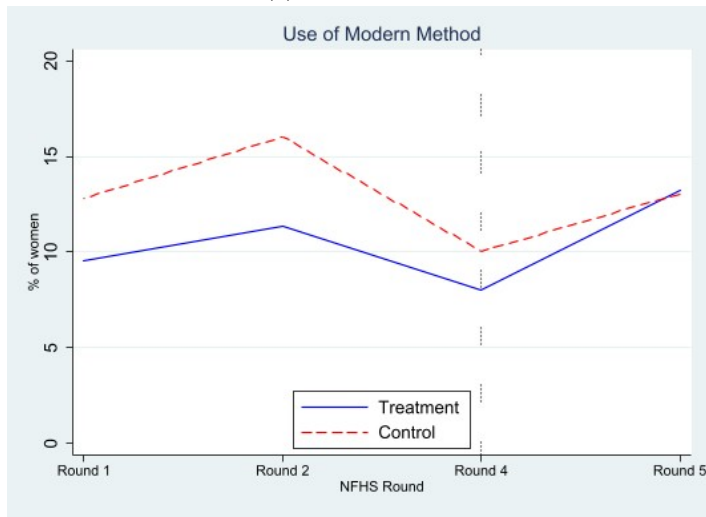
This section tests for parallel trends for the difference-in-differences approach. In Figure 3, graph (a) displays the average proportion of respondent who have heard of HIV for all NFHS rounds for both the treatment and control groups. The graph shows that awareness of HIV between treatment and control groups follows parallel trends pre-intervention, with convergence post-intervention, indicating the program's potential impact. Similarly, the graph (b) shows trends in knowledge of modern methods of contraception across NFHS rounds. Before the intervention, the treatment and control groups follow parallel trends, reinforcing the assumption of similar pre-treatment behavior. Post-intervention, there is a clear convergence between the groups, with the average for the treatment group surpassing that of the control group.



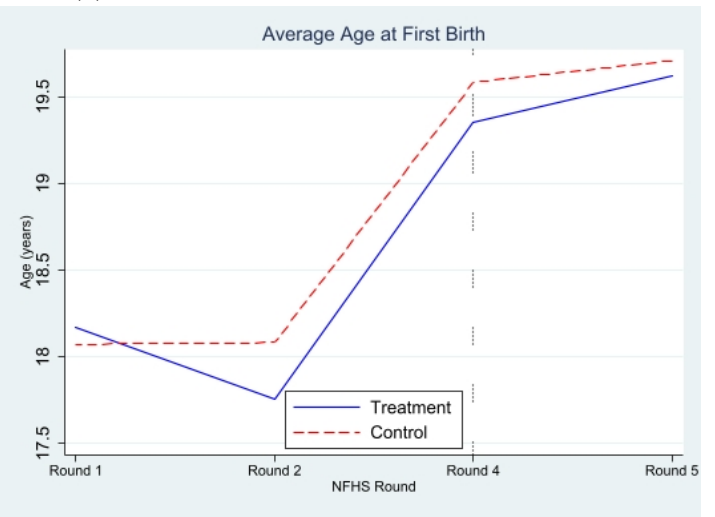
(a) Heard of HIV



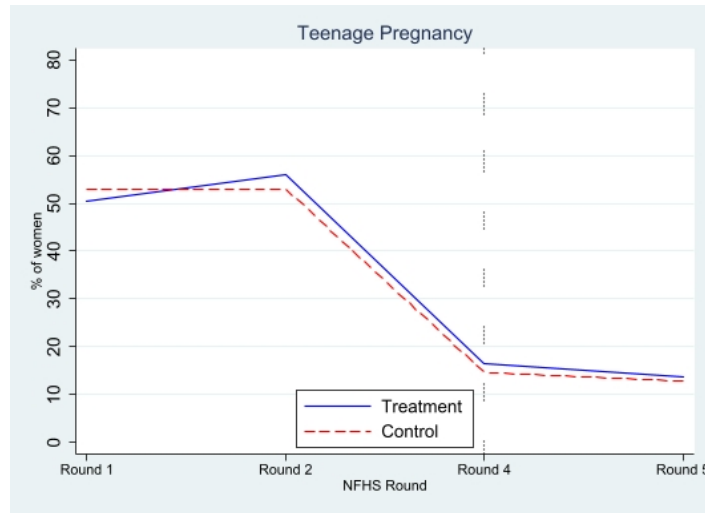
(b) Knowledge of modern methods of contraception



(c) Use of modern methods of contraception



(d) Age of respondent at first birth



(e) Teenage pregnancy

Figure 3: Horizontal axis represent NFHS rounds NFHS-1: 1992-93, NFHS-2: 1998-99, NFHS-4: 2015-16, NFHS-5: 2019-21. The dashed line on the graph separates the pre-treatment and post-treatment periods.

Next, the graph (c) on modern contraceptive use shows similar pre-intervention trends, with the treatment group ex-

periencing a notable post-intervention rise. The graph (d) on age at first birth also reveals parallel trends pre-intervention, with the gap narrowing post-intervention. Finally, the teenage pregnancy highlights a significant declining trend in teenage pregnancies, with over 50% of girls aged 15-25 having their first birth before age 20 in the 1990s, dropping to around 10% in more recent years. Although the treatment group consistently had higher rates of teenage pregnancies compared to the control group, the post-intervention period shows a narrowing of this gap, indicating some convergence between the two groups over time.

7 Results

In this section, I present the diff-in-diff results for each outcome variable.

7.1 Knowledge of HIV

In this study, two variables are used to measure HIV knowledge: (1) Heard of HIV, based on whether respondents have heard of HIV/AIDS, and (2) Comprehensive Knowledge of HIV, constructed from responses to five questions related to HIV transmission and misconceptions. These include: "Can you reduce the risk of getting HIV by always using condoms during sex?" "Can you reduce the risk of getting HIV by having only one sex partner with no other partners?" "Can you get HIV from mosquito bites?" "Can you get HIV by sharing food with a person who has AIDS?" and "Can a healthy-looking person have HIV?" The outcome variable is set to 1 if all questions are answered correctly. This differs from the NFHS report, which includes additional questions. ¹⁷

VARIABLES	Heard of HIV				Comprehensive knowledge of HIV	
	Sample: All years		Sample:NFHS-4,5		Sample:NFHS-4,5	
Treatment * Post	0.130*** (0.0241)	0.128*** (0.0239)	0.119*** (0.0240)	0.117*** (0.0238)	0.0195 (0.0177)	0.0201 (0.0177)
Treatment	-0.123*** (0.0160)	-0.108*** (0.0151)	-0.110*** (0.0179)	-0.0992*** (0.0169)	-0.0369*** (0.0141)	-0.0314** (0.0139)
Constant	0.399*** (0.0277)	0.328*** (0.0315)	0.847*** (0.0246)	0.741*** (0.0297)	0.112*** (0.0161)	0.0240 (0.0226)
Observations	62,710	62,710	48,446	48,446	38,907	38,907
R-squared	0.263	0.284	0.100	0.117	0.043	0.050
Sample Mean	.704 (0.0018)		.803 (0.0018)		.252 (0.0022)	.
Controls	NO	YES	NO	YES	NO	YES
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 2: Knowledge of HIV

Table 2 reports the estimates of the program effects on raising awareness regarding HIV/AIDS. The outcome here is "Heard of HIV", that is equals to 1 if respondents answers 'Yes'. The first column, which includes all NFHS rounds, shows a positive and significant impact of the program. The regression estimate suggests that the program increases the likelihood that a respondent is aware of HIV/AIDS by 13 percentage points if she is in the treated district. Column 2 adds controls for religion, area of residence, and age, and the results remain robust to these additional controls with 12.8

¹⁷NFHS Round 1 (1992-93) does not include any questions on HIV

percentage point increase. In columns 3 and 4 using data from NFHS-4 (pre-treatment) and NFHS-5 (post-treatment), the coefficient decreases but still indicates a positive effect, with an 11.7 percentage point increase in HIV awareness.

Columns 5 and 6 reports the results for comprehensive knowledge of HIV. The findings indicate that while the effects are positive, they are not statistically significant. To gain further insight, I run regressions for each individual question. Table 8 in the appendix presents the results for each question separately. Of all the questions, only the one regarding reducing the risk of HIV by having one sex partner who does not have other partners shows a significant coefficient, while the others do not. Overall, the results in Table 8 suggest some positive effects of the program on improving knowledge of HIV for treated districts, but these effects are not statistically significant. Hence, the results for knowledge of HIV variable suggests that program has some positive effect on the extensive margin; however, it is not that effective on the intensive margin. The program effectively increased the number of young girls who heard of HIV however, it did not result in more girls who understands the ways to avoid HIV and what are common misconception regarding HIV.¹⁸

7.2 Knowledge and use of modern method of contraception

This section focuses on the knowledge and use of modern methods of contraception. Here, the outcome variable is equal to 1 if the respondent answers that they know modern methods of contraception. The corresponding diff-in-diff results are reported in Table 3. The first four columns present the results for knowledge of modern contraception methods. Columns 1 and 2, which include all NFHS rounds, show a positive and significant impact of the program. Columns 3 and 4, using data from NFHS-4 (pre-treatment) and NFHS-5 (post-treatment), continue to show positive effects. Columns 2 and 4 incorporate controls for religion, area of residence, and age, with the results remaining robust to these additions. The estimated impact across these columns ranges from a 1.25 to 1.36 percentage points increase, indicating that the program improved adolescent girls' knowledge of modern contraception by approximately 1.3 percentage points.

VARIABLES	Knowledge of Method				Use of Method			
	Sample: All years		Sample:NFHS-4,5		Sample: All years		Sample:NFHS-4,5	
Treatment * Post	0.0131** (0.00566)	0.0136** (0.00567)	0.0125** (0.00597)	0.0130** (0.00598)	0.0241*** (0.00664)	0.0263*** (0.00645)	0.0210*** (0.00621)	0.0233*** (0.00607)
Treatment	-0.0129*** (0.00433)	-0.0114*** (0.00425)	-0.0123** (0.00484)	-0.0109** (0.00477)	-0.0174*** (0.00414)	-0.0160*** (0.00393)	-0.0144*** (0.00434)	-0.0136*** (0.00407)
Constant	0.899*** (0.0130)	0.759*** (0.0137)	0.959*** (0.00467)	0.826*** (0.00727)	0.186*** (0.0115)	-0.498*** (0.0160)	0.156*** (0.00981)	-0.498*** (0.0138)
Observations	338,135	338,135	309,019	309,019	338,135	338,135	309,019	309,019
R-squared	0.008	0.026	0.004	0.023	0.012	0.109	0.012	0.114
Sample Mean	0.970 (0.0002)		0.972 (0.0002)		0.112 (0.0005)		0.112 (0.0005)	
Controls	NO	YES	NO	YES	NO	YES	NO	YES
State FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 3: Knowledge and use of modern methods of contraception

Table 3, columns 5 to 8 present the results for the use of modern methods of contraception. The table reveals that the intervention led to an increase of 2.33 percentage points in the use of modern contraceptives in the treatment group. The coefficients for different specifications are quite similar, suggesting consistent effects of the intervention across various model setups. The "Treatment" variable alone indicates reduced contraceptive use before the intervention in treatment districts relative to control districts. This is not surprising given that these are categorized as HPDs.

¹⁸For consistency data is only from NFHS-4 and NFHS-5

7.3 Teenage pregnancy

The outcome variables in this section is the age of the respondent at first birth and teenage pregnancy. As the program module covers consequences of early marriage and teenage pregnancy, these outcomes help to understand the effects of the program on adolescent parents and if the program is working toward increasing the age of women at first birth/decreasing early child bearing.

VARIABLES	Age of Mother				Teenage Pregnancy			
	Sample: All years		Sample:NFHS-4,5		Sample: All years		Sample:NFHS-4,5	
Treatment * Post	0.0896*	0.0830*	0.121**	0.104**	-0.0201**	-0.0189**	-0.0278***	-0.0245***
	(0.0519)	(0.0442)	(0.0497)	(0.0436)	(0.00970)	(0.00838)	(0.00977)	(0.00866)
Treatment	-0.235***	-0.161***	-0.268***	-0.185***	0.0395***	0.0262***	0.0494***	0.0337***
	(0.0538)	(0.0442)	(0.0548)	(0.0448)	(0.00873)	(0.00709)	(0.00988)	(0.00806)
Constant	17.32***	9.544***	18.76***	10.43***	0.852***	2.291***	0.612***	2.238***
	(0.101)	(0.119)	(0.0834)	(0.127)	(0.0158)	(0.0222)	(0.0157)	(0.0268)
Observations	112,079	112,079	90,872	90,872	112,079	112,079	90,872	90,872
R-squared	0.099	0.205	0.039	0.146	0.062	0.146	0.028	0.113
Sample Mean	19.30		19.59		.528	.	.482	.
	(0.0071)		(0.0076)		(.0014)		(0.0016)	
Controls	NO	YES	NO	YES	NO	YES	NO	YES
State FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 4: Teenage Pregnancy

Columns 1 through 4 in Table 4 show the results for the age of the respondent at the first birth. The main result that adds controls for religion, area of residence, and age suggests that the intervention led to a significant increase in the age at first birth, with the treatment group experiencing an increase of 0.0830 to 0.121 years after the intervention. Regarding teenage pregnancy, this outcome variable calculates the proportion of women who gave birth before the age of 20, given that they have children. Hence, it represents the proportion of respondents who were in their teens at the time of their first birth.¹⁹ The columns 5 and 6 in Table 4, which includes data from all NFHS rounds, show a statistically significant negative coefficient, that shows the program has reduced teenage pregnancy by 1.84 percentage points. Columns 7 and 8 focus on NFHS-4 (pre-treatment) and NFHS-5 (post-treatment). Column 8 adds controls for religion, area of residence, and age, and the results remain robust to these additional controls, showing even greater decline compared to column 5. The main result of this analysis suggests that the program effectively reduced teen pregnancy by 2.45 percentage points.

Discussion: For outcomes related to teenage pregnancy and childbirth, the validity of the identifying assumptions may be limited. Since High Priority Districts are selected based on a range of maternal health indicators, it is likely that other programs and benefits targeted at these districts are also driving the observed improvements. Therefore, while the Peer Educator program may have played a role, positive changes cannot be solely attributed to it, as other interventions may have also contributed significantly to these outcomes.

7.4 Attitude towards gender based violence

To assess the role and effectiveness of program in changing attitude towards gender based violence, I examined the outcome related to 'justifying wife beating'. The NFHS questionnaire includes a series of scenarios in which respondents are asked whether they believe a husband is justified in beating his wife. The outcome variable equals 1 when a respondent considers

¹⁹For this analysis, the Teen pregnancy variable does not include the "currently pregnant" respondents

wife beating justified under any of the given circumstances. On average, about 40 percent of women aged 15-25 in the NFHS-4 and NFHS-5 samples justify wife beating in at least one circumstance. The negative coefficients indicate that the program reduces the share of women who believe wife beating is justified. However, these results are not statistically significant.

VARIABLES	Justified wife beating Sample:NFHS-4,5	
Treatment * Post	-0.00509 (0.0219)	-0.00292 (0.0218)
Treatment	0.0382** (0.0162)	0.0287* (0.0158)
Constant	0.458*** (0.0325)	0.334*** (0.0360)
Observations	48,446	48,446
R-squared	0.075	0.082
Sample Mean	0.406 (0.0022)	
Controls	NO	YES
State FE	YES	YES
Year FE	YES	YES
Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1		

Table 5: Justifying wife beating

8 Mechanisms

To explore the mechanisms through which the program influences the outcome variables, I analyzed relevant data from the NFHS survey, focusing on marital status, sexual activity, age at marriage, and indicators of women's autonomy within marriage. The statistics show that 96.2% of never-married women in the treatment districts reported not engaging in sexual activity, compared to 97.3% in the control districts. These findings suggest that engagement in risky behaviors may largely stem from early marriage rather than premarital sexual activity, underlining the importance of addressing early marriage as a critical driver of adolescent sexual behavior.

To evaluate the program's effect on early marriage, a key determinant of adolescent sexual activity, I analyzed the age at first marriage. In treatment districts, the average age at marriage increased significantly from 17.68 years in NFHS-4 to 18.16 years in NFHS-5, while in control districts, it rose from 18.07 to 18.36 years. The regression results suggests a significant treatment effect, indicating that the program successfully contributed to delaying marriage in treatment districts relative to control districts. However, this increase in the age at marriage did not translate into a significant change in the age at first sexual activity. This finding implies that while the program has some impact on early marriage, its influence on reducing engagement in risky sexual behaviors remains limited.

Additionally, I examined women's knowledge of marital rights and their bargaining power within marriage, which are critical dimensions of autonomy. When asked if a wife is justified in refusing sex if her husband has a sexually transmitted disease, 74.34% of women in treatment districts and 75.30% in control districts responded affirmatively. This minimal difference suggests that the program has had little impact on improving women's autonomy or awareness of their marital rights.²⁰ Overall, these findings underscore the program's effectiveness in delaying marriage but highlight its constrained impact on addressing risky sexual behaviors and enhancing women's autonomy, suggesting the need for a more comprehensive approach to adolescent reproductive health.

²⁰The comparable data from NFHS-4 is not available, which limits the ability to assess changes over time for this variable

9 Robustness Check

As robustness check to further explore if there are pre-trends, I use the previous NFHS rounds as placebo treatment periods. Table 6 reports the results for the outcome variables with NFHS round 1 and NFHS round 2 as placebo treatment periods. The insignificant coefficient support the robustness of the main results suggesting that the parallel trend assumption holds.²¹

VARIABLES	Heard of HIV	Knowledge of modern methods		Use of Method	
NFHS 2* Treatment	-0.0159 (0.0303)		-0.00366 (0.00823)		-0.0254** (0.0129)
Treatment	-0.0955*** (0.0166)	-0.0120*** (0.00443)	-0.0120*** (0.00453)	-0.0153*** (0.00388)	-0.0139*** (0.00387)
NFHS 1* Treatment		0.00107 (0.0214)		-0.0107 (0.0136)	
Constant	0.768*** (0.0287)	0.848*** (0.00735)	0.848*** (0.00736)	-0.412*** (0.0150)	-0.440*** (0.0168)
Observations	41,151	193,643	193,643	193,643	193,643
R-squared	0.292	0.027	0.027	0.098	0.098
State FE	YES	YES	YES	YES	YES
NFHS Year FE	YES	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 6: Robustness Check

Next, I re-estimated the results using district fixed effects to account for unobserved heterogeneity across districts that might influence the outcomes. The results in table 7 suggests that the main results remain robust to this specification, with the coefficients for all outcome variables maintaining both their magnitude and direction. This consistency reinforces the validity of the results and suggests that the observed program effects are not driven by district-specific factors.

VARIABLES	Heard of HIV	Knowledge of HIV	Knowledge of method	Use of method
Treatment * Post	0.118*** (0.0245)	0.0121 (0.0182)	0.0112* (0.00605)	0.0216*** (0.00607)
treatment	0.00365 (0.0164)	-0.0923*** (0.0138)	-0.113*** (0.00409)	-0.100*** (0.00465)
Constant	0.701*** (0.0238)	0.181*** (0.0213)	0.859*** (0.00670)	-0.498*** (0.0118)
Observations	48,446	38,907	309,019	309,019
R-squared	0.154	0.073	0.037	0.122
State FE	YES	YES	YES	YES
District FE	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

²¹I have taken the NFHS round and not year as Treatment period hence taken NFHS round fixed effects the results are same for year fixed effects

VARIABLES	Age of Mother	Teen Pregnancy	Justified DV
Treatment * Post	0.0839** (0.0419)	-0.0214** (0.00852)	-0.0122 (0.0221)
Treatment	0.881*** (0.0299)	-0.0765*** (0.00619)	0.321*** (0.0163)
Constant	10.49*** (0.115)	2.208*** (0.0215)	0.275*** (0.0233)
Observations	90,872	90,872	48,446
R-squared	0.164	0.127	0.118
State FE	YES	YES	YES
District FE	YES	YES	YES
Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1			

Table 7: District FE

10 Conclusion

This study evaluates the impact of the Peer Educator program under the Rashtriya Kishor Swasthya Karyakram (RKSK) on various outcomes, including knowledge of HIV, use of modern methods of contraception, teenage pregnancy rates, and attitudes toward gender-based violence. By employing a difference-in-differences framework, the research compares high-priority districts where the program was implemented with other districts that did not receive the intervention in the state.

The empirical analysis suggests that while the program has been effective in increasing general awareness about HIV and improving both knowledge and use of modern contraception, its impact on more comprehensive knowledge of HIV and changes in attitudes towards gender-based violence has been limited. The program also shows a significant reduction in teenage pregnancy rates and an increase in age at first birth. Overall, the study highlights the effectiveness of SRH education in empowering adolescents but also points out the need for more comprehensive approaches to tackle deeper issues related to adolescent health and well-being.

The program shows potential but requires improvements. According to Arora et al. (2024), peer educators face challenges such as high dropout rates due to marriage, relocation, and inadequate incentives. Parental concerns about academic performance and lack of adolescent motivation further hinder participation, while limited resources and scheduling conflicts also pose significant barriers for adolescents.

Furthermore, the period considered in our analysis coincided with the COVID-19 pandemic, during which peer educator meetings faced significant disruptions. However, peer educators played an important role in supporting healthcare workers during this time. Bassi et al. (2023) report that peer educators contributed to pandemic response efforts, addressed community and adolescent needs, assisted in vaccination drives, and helped implement national health programs. These findings suggest that by improving meeting facilitation and improving incentives and improved incentives, the Peer Educator program could achieve even better health outcomes.

The study primarily addresses broader outcomes, but exploring the underlying mechanisms in greater detail could enhance program evaluation and offer valuable insights. Additionally, while this paper focuses on the impact of the program on adolescent girls, expanding the sample to include all adolescents could provide a more comprehensive understanding of the program’s overall effectiveness and its potential to challenge gender norms.

11 Appendix

11.1 Comprehensive Knowledge by questions

VARIABLES	Reduce risk:		Can get HIV		Healthy looking person can have AIDs
	Always use condoms during sex	Have 1 sex partner	From mosquito bites	By sharing food with person who has aids	
Treatment * Post	0.0298* (0.0164)	0.0288* (0.0156)	0.00470 (0.0185)	0.0116 (0.0177)	0.0130 (0.0145)
Treatment	-0.0239** (0.0120)	-0.0330*** (0.0119)	-0.0274** (0.0135)	-0.0200 (0.0130)	-0.0137 (0.0118)
Constant	0.302*** (0.0342)	0.283*** (0.0304)	0.509*** (0.0391)	0.395*** (0.0397)	0.669*** (0.0398)
Observations	38,907	38,907	38,907	38,907	38,907
R-squared	0.030	0.041	0.029	0.054	0.026
Controls	YES	YES	YES	YES	YES
State FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 8: Outcome: Comprehensive knowledge of HIV (by question)

References

- Monika Arora, Shalini Bassi, Deepika Bahl, and Stefanie Dringus. 129. facilitators and barriers influencing the implementation of peer education programme under national adolescent health strategy: Findings from i-saathiya study. *Journal of Adolescent Health*, 74(3):S69–S70, 2024.
- Oriana Bandiera, Niklas Buehren, Robin Burgess, Markus Goldstein, Selim Gulesci, Imran Rasul, and Munshi Sulaiman. Women’s empowerment in action: evidence from a randomized control trial in africa. *American Economic Journal: Applied Economics*, 12(1):210–259, 2020.
- Shalini Bassi, Deepika Bahl, Heeya Maity, Stefanie Dringus, and Monika Arora. 52. engagement of peer educators from india’s national adolescent health programme during the covid-19 pandemic response. *Journal of Adolescent Health*, 72(3):S33–S34, 2023.
- Nina Buchmann, Erica Field, Rachel Glennerster, Shahana Nazneen, Svetlana Pimkina, and Iman Sen. Power vs money: Alternative approaches to reducing child marriage in bangladesh, a randomized control trial. *Abdul Latif Jameel Poverty Action Lab (J-PAL) working paper*, 2018.
- Prerna Dewan, Tridip Ray, Arka Roy Chaudhuri, and Kirti Tater. Gender peer effects in high schools: Evidence from india. *Journal of Economic Behavior & Organization*, 220:470–494, 2024.
- Diva Dhar, Tarun Jain, and Seema Jayachandran. Reshaping adolescents’ gender attitudes: Evidence from a school-based experiment in india. *American economic review*, 112(3):899–927, 2022.
- Thomas J Dishion and Jessica M Tipsord. Peer contagion in child and adolescent social and emotional development. *Annual review of psychology*, 62(1):189–214, 2011.
- Esther Duflo, Pascaline Dupas, and Michael Kremer. Education, hiv, and early fertility: Experimental evidence from kenya. *American Economic Review*, 105(9):2757–2797, 2015.

- Pascaline Dupas. Do teenagers respond to hiv risk information? evidence from a field experiment in kenya. *American Economic Journal: Applied Economics*, 3(1):1–34, 2011.
- Pascaline Dupas, Elise Huillery, and Juliette Seban. Risk information, risk salience, and adolescent sexual behavior: Experimental evidence from cameroon. *Journal of Economic Behavior & Organization*, 145:151–175, 2018.
- Eric Edmonds, Ben Feigenberg, and Jessica Leight. Advancing the agency of adolescent girls. *Review of Economics and Statistics*, 105(4):852–866, 2023.
- James J Heckman and Stefano Mosso. The economics of human development and social mobility. *Annu. Rev. Econ.*, 6(1):689–733, 2014.
- Seema Jayachandran. The roots of gender inequality in developing countries. *Annual review of economics*, 7(1):63–88, 2015.